■# PAPER_243 — NGC 3603 Full MUGE: Time-Varying Mass M(t) and Additive Cavity Pressure P(t)/ρ

Author: Daniel T. Murphy

Version: *1.0* | ***Session:*** *60* | ***PAPER Number:*** *243*

■# PAPER_243 — NGC 3603 Full MUGE: Time-Varying Mass M(t) and Additive Cavity Pressure P(t)/ρ

NGC 3603 Extreme Young Star Cluster — Complete 10-Term Master Universal Gravity Equation

Author: Daniel T. Murphy **Framework:** Unified Quantum Field Framework (UQFF) v4.10 **Calculator Class:** `NGC3603FullMUGECavityPressureCalculator` (CP3, Session 60) **Encoded By:** Grok (xAI), October 2025 (C++ source); Python CP3 integration March 2026 **Version:** 1.0 | **Session:** 60 | **PAPER Number:** 243

<!-- UQFF constants: κ = 5.0e-4 day^{-1}, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

This paper presents the complete 10-term Master Universal Gravity Equation (MUGE) for the NGC 3603 extreme young star cluster, incorporating two novel mathematical elements not previously captured in CP3:

Time-varying mass `M(t) = M_0(1 + \dot{M}_{\rm factor}\,e^{-t/\tau_{SF}})` — exponential

star-formation inflow driving cluster mass growth over the first few Myr.

Additive cavity pressure acceleration `P(t)/\rho_{\rm fluid}` where

`P(t) = P_0\,e^{-t/\tau_{\rm exp}}` — the ionized cavity pressure from O/B-star UV/wind feedback expressed as an acceleration term alongside (not as a modifier of) the gravitational field.

These are **expressly distinct** from CP3 class 88 (`NGC3603StellarPressureModulationCalculator`, Session 55), which uses pressure as a multiplicative suppressor `g \propto (1 - P(t))`. The present paper treats pressure as an independent additive acceleration channel.

---

2. System Parameters

Parameter	Symbol	Default Value	Units
Initial mass	`M_0`	`4\times10^5\,M_\odot`	kg
Cluster radius	`r`	`9.5\,\rm ly`	m
Star-formation timescale	`\tau_{\rm SF}`	`1\,\rm Myr`	s
Initial cavity pressure	`P_0`	`4\times10^{-8}`	Pa
Pressure expansion timescale	`\tau_{\rm exp}`	`1\,\rm Myr`	s
Wind density	`\rho_{\rm wind}`	`10^{-20}`	kg/m^3
Wind velocity	`v_{\rm wind}`	`2\times10^6`	m/s
Fluid density	`\rho_{\rm fluid}`	`10^{-20}`	kg/m^3

Parameter	Symbol	Default Value	Units
Magnetic field	B	10^{-5}	T

---

3. Novel Mathematical Elements

3.1 Time-Varying Cluster Mass

$$M(t) = M_0\!\left(1 + \dot{M}_{\rm factor}\,e^{-t/\tau_{SF}}\right)$$

where $\dot{M}_{\rm factor}$ is the dimensionless peak accretion enhancement. At $t=0$, $M(0) = M_0(1 + \dot{M}_{\rm factor})$ (maximum infall); at $t \gg \tau_{SF}$, $M \to M_0$ (steady state). The **star-formation rate** driving this is:

$$\frac{dM}{dt} = -\frac{M_0\,\dot{M}_{\rm factor}}{\tau_{SF}}\,e^{-t/\tau_{SF}}$$

The **star-formation efficiency** at time t is:

$$\varepsilon_{SF}(t) = \frac{M(t) - M_0}{M_0} = \dot{M}_{\rm factor}\,e^{-t/\tau_{SF}}$$

3.2 Additive Cavity Pressure Acceleration

The O/B-star population inflates an ionized HII cavity. The cavity pressure decays exponentially as the molecular cloud is dispersed:

$$P(t) = P_0\,e^{-t/\tau_{\rm exp}}$$

This converts to a specific acceleration (per unit mass of ambient gas):

$$T_{\rm pressure} = \frac{P(t)}{\rho_{\rm fluid}} = \frac{P_0\,e^{-t/\tau_{\rm exp}}}{\rho_{\rm fluid}}$$

Dispersal condition: The cluster disperses its natal cloud when the cavity-pressure acceleration exceeds local gravity:

$$\frac{P(t_{\rm disp})}{\rho_{\rm fluid}} = T_1 \Rightarrow t_{\rm disp} = \tau_{\rm exp}\ln\!\left(\frac{P_0}{\rho_{\rm fluid} \cdot T_1}\right)$$

---

4. Full 10-Term MUGE

$$g_{\rm NGC3603}(r,t) = T_1 + T_2 + T_3 + T_4 + T_q + T_{\rm fl} + T_{\rm osc} + T_{\rm DM} + T_{\rm wind} + T_{\rm pressure}$$

Term 1 — Base with M(t), Hubble, magnetic:

$$T_1 = \frac{GM(t)}{r^2}(1 + H_0\,t)(1 - B/B_{\rm crit})$$

Note: Unlike class 88 which applies $(1-P)$ here, the full MUGE leaves T■ unmodified and adds P explicitly as T■■.

Term 2 — UQFF unified field:

$$T_2 = (U_{g1}(t) + U_{g4}(t))(1 + f_{TRZ}), \quad U_{gk}(t) \propto M(t)/r^2$$

Term 3 — Cosmological constant:

$$T_3 = \frac{\Lambda c^2}{3}$$

Term 4 — EM with vacuum ratio:

$$T_4 = \frac{qv_{\rm gas}B}{m_p}\!\left(1 + \frac{\rho_{\rm UA}}{\rho_{\rm SCm}}\right)\! s_{\rm EM}$$

Term 5 — Quantum uncertainty:

$$T_q = \frac{\hbar}{\sqrt{\Delta x\,\Delta p}}\,\psi\,\frac{2\pi}{t_H}$$

Term 6 — Fluid acceleration:

$$T_{\rm fl} = \frac{\rho_{\rm fluid}\,V\,g_{\rm base}(t)}{M(t)}$$

Term 7 — Two-mode oscillation:

$$T_{\rm osc} = 2A\cos(kx)\cos(\omega t) + \frac{2\pi}{t_{\rm Gyr}}A\cos(kx - \omega t)$$

Term 8 — Dark matter with $3GM(t)/r^3$ tidal perturbation:

$$T_{\rm DM} = \frac{(M(t) + M_{\rm DM})\!\left(\delta\rho/\rho + 3GM(t)/r^3\right)}{M(t)}$$

Term 9 — Stellar wind ram pressure:

$$T_{\rm wind} = \frac{\rho_{\rm wind}\,v_{\rm wind}^2}{\rho_{\rm fluid}}$$

Term 10 — Cavity pressure (additive, novel):

$$T_{\rm pressure} = \frac{P_0\,e^{-t/\tau_{\rm exp}}}{\rho_{\rm fluid}}$$

---

5. Distinction from Prior Art

Feature	Class 88 (Session 55)	This class (Session 60)
Mass	Fixed M	$M(t) = M_0(1 + \dot{M}e^{-t/\tau})$
Pressure form	Multiplicative $(1-P(t))$ modifier	Additive $P(t)/\rho_{\rm fluid}$ term
Physics	UV/wind pressure suppresses gravity	Cavity pressure drives independent acc.
Terms	4 (base, Ug, Λ, wind)	10 (all MUGE terms)
Quantum term	Not present	T_q present
Fluid term	Not present	$T_{\rm fl}$ present
2-mode oscillation	Not present	Standing + traveling wave
DM pert2	Not present	$3GM(t)/r^3$

---

6. Cross-Validation: Class 88 vs Class 111 Regime

For small P_0 and short $t \ll \tau_{\rm exp}$:

$$P(t) \approx P_0 \Rightarrow T_{\rm pressure} \approx P_0/\rho_{\rm fl}$$

For Class 88, the $(1-P)$ effect on T_1:

$$\Delta T_1 = -T_1^{\rm Class88} \cdot P \approx -\frac{GM}{r^2} P$$

The two are **equivalent only in the limit** where $P_0/\rho_{\rm fl} = (GM/r^2) P$, i.e., when $P_0 = GM\rho_{\rm fl}/r^2$. At default values of NGC 3603, this equality does not hold — the additive and multiplicative forms predict **different total** g and different dispersal timescales.

---

7. Numerical Example

At $t = 0.5\,\rm Myr$, default parameters:

$$M_{\rm dot}(t) = 1.0 \times e^{-0.5} \approx 0.607 \Rightarrow M(t) \approx 1.607\,M_0$$

$$P(t) = 4\times10^{-8} \times e^{-0.5} \approx 2.43\times10^{-8}\,\rm Pa$$

$$T_{\rm pressure} = \frac{2.43\times10^{-8}}{10^{-20}} = 2.43\times10^{12}\,\rm m/s^2$$

The cavity pressure acceleration dominates all other terms at early times — consistent with the observational picture of NGC 3603's OB-star cluster rapidly dispersing its natal cloud within $\sim3\,\rm Myr$ (Harayama et al. 2008; Pang et al. 2013).

---

8. Simulation Recommendations

- τ_{SF} **sweep** (0.5–5 Myr): Track $M(t)$ peak mass vs. cluster binding energy
- P_0 **sweep** (10^{-9}–10^{-6} Pa): Determine cavity pressure dispersal threshold
- **Term10 vs Term9 cross-over**: Find t^* where $T_{\rm pressure}(t^*) = T_{\rm wind}$
- $M_{\rm dot}$ **sweep** (0.1–2.0): Map star-formation efficiency vs. gravitational retention

---

9. Integration Into UQFF Pipeline

- **CP3 class:** `NGC3603FullMUGECavityPressureCalculator` (113th AST node, 112th calculator, Session 60)
- **Aggregator:** v2.4.0 — registered in `CP3_CALCULATORS`
- **Source:** Doc 11 C++ class `NGC3603` (Grok/xAI, October 2025)
- **Complementary class:**
- Class 88 `NGC3603StellarPressureModulationCalculator` — multiplicative pressure form (Session 55)
- Class 110 `RingsOfRelativityEinsteinLensingMUGECalculator` — companion new class (PAPER_242)

---

10. References

- Harayama Y. et al. (2008): ApJ 675, 1319 — NGC 3603 stellar mass function
- Pang X. et al. (2013): ApJ 764, 73 — star formation history
- Portegies Zwart S. et al. (2010): Nature 464, 203 — young massive clusters
- Murphy, D.T. (2025): UQFF Manuscript — Doc 11 (NGC 3603 MUGE, October 2025)
- Grok (xAI): C++ class `NGC3603`, encoded October 2025

PAPER_243 | Session 60 | CP3 class 112 (NGC3603FullMUGECavityPressureCalculator) | UQFF v4.10
